Applying Machine Learning Techniques in Recognizing and Evaluating Student Feedback

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Abstract

In Vietnam's rapidly evolving higher education landscape, analyzing student feedback and evaluating teachers at the end of courses is essential to transform teaching quality and personalizing instructional strategies. This study tackles the challenge of classifying 16,175 feedback entries from the Vietnam Student Feedback Data Set on Hugging Face into three sentiment categories: Positive, Negative, and Neutral. We rigorously compare traditional machine learning models Logistic Regression, SVM, and Naive Bayes with cutting-edge approaches, including XGBoost, a stacking ensemble, and PhoBERT, a BERT-based model fine-tuned for Vietnamese. Evaluated on accuracy, precision, recall, and F1-score, PhoBERT demonstrates exceptional ability to capture nuanced sentiments, with Stacking also delivering robust results. This research empowers institutions with automated and precise feedback analysis, driving impactful improvements in teaching effectiveness and student satisfaction in Vietnam.

1 Introduction

The rapid growth of digital platforms in education has led to an influx of student feedback, which requires effective sentiment analysis to improve teaching quality and institutional decision-making. Classifying student feedback into three categories: Positive, Negative, and Neutral presents challenges due to the linguistic complexity of the Vietnamese language, including tonal nuances and contextual ambiguities. Additionally, the brevity of feedback texts and the limited availability of annotated datasets further complicate this task. Traditional machine learning methods, such as Logistic Regression, Support Vector Machines (SVM) (Manning et al., 2008), and Naive Bayes (Jurafsky and Martin, 2023), often struggle to capture semantic subtleties. In contrast, advanced approaches like XGBoost (Chen and Guestrin, 2016), Stacking ensemble methods, and PhoBERT, a BERT-based model pre-trained for Vietnamese (Nguyen et al., 2020), excel in modeling complex linguistic patterns.

This study proposes a robust framework for classifying 16,175 Vietnamese student feedback instances, combining data collected from educational platforms with synthetically generated feedback using large language models. We systematically compare traditional machine learning approaches (Logistic Regression, SVM, Naive Bayes) with advanced methods (XGBoost, Stacking ensemble, and PhoBERT) to determine the most effective solution. Models are evaluated based on accuracy, precision, recall, and F1-score, addressing linguistic and data challenges. Our contributions include a comprehensively annotated dataset, a rigorous comparative analysis, and insights into the efficacy of advanced models, particularly PhoBERT, for Vietnamese sentiment analysis, advancing feedback evaluation and educational improvement.

Contributions. The primary contributions of this study are:

- A comprehensive framework for Vietnamese student feedback classification, systematically integrating traditional machine learning (Logistic Regression, SVM, Naive Bayes) with advanced approaches (XGBoost, Stacking ensemble, PhoBERT).

- A curated dataset of 16,175 annotated Vietnamese feedback instances, combining collected and synthetically generated data, addressing the scarcity of labeled resources for Vietnamese NLP.

- An in-depth comparative analysis of model performance, highlighting the strengths of advanced models, particularly PhoBERT, for enhancing sentiment analysis and supporting educational decision-making.

2 Related Work

Text classification, particularly sentiment analysis, has been extensively studied in natural language processing, with traditional machine learning methods achieving notable success on English datasets. Logistic Regression and Support Vector Machines (SVM) (Manning et al., 2008) are computationally efficient and perform well on high-dimensional, sparse data, but struggle to capture semantic nuances due to their reliance on term frequency-based features like TF-IDF. Naive Bayes (Jurafsky and Martin, 2023) offers simplicity and speed through probabilistic inference, but its assumption of feature independence often results in suboptimal performance on complex texts. Early Vietnamese NLP studies applied these methods to sentiment analysis tasks, but their effectiveness was limited by the tonal complexity of the Vietnamese language and the scarcity of annotated datasets (Jurafsky and Martin, 2023).

Advanced machine learning approaches have significantly improved Vietnamese text classification. XGBoost (Chen and Guestrin, 2016), a gradient-boosting framework, excels in handling structured data and capturing non-linear relationships, but may require careful feature engineering for text tasks. Stacking ensemble methods combine multiple models to enhance predictive performance, leveraging the strengths of individual classifiers, though they increase computational complexity. PhoBERT (Nguyen et al., 2020), a BERT-based model pre-trained on Vietnamese corpora, utilizes transformer-based contextual embeddings, outperforming traditional models in tasks like sentiment analysis and named entity recognition. However, PhoBERT demands significant computational resources for fine-tuning. Previous studies often relied on small, domain-specific datasets and focused on single-model evaluations, lacking comprehensive comparisons. Our work addresses these gaps with a 16,175 annotated Vietnamese student feedback dataset, combining collected and synthetically generated data, and a systematic evaluation of Logistic Regression, SVM, Naive Bayes, XGBoost, Stacking ensemble, and PhoBERT, highlighting PhoBERT's superior contextual understanding for Vietnamese sentiment analysis.

3 Methodology

Text classification is a core task in natural language processing (NLP), essential for analyzing digital content, such as classifying Vietnamese student feedback into sentiment categories: Positive, Negative, and Neutral. Below, we provide a comprehensive review of the theoretical foundations, strengths, and weaknesses of the key methods and models used in this study, focusing on their applicability to Vietnamese NLP. Detailed formulas, explanations, and examples are included, tailored to sentiment analysis of student feedback.

3.1 Bag-of-Words (BoW)

The **Bag-of-Words (BoW)** model (Manning et al., 2008) represents text as a vector of word frequencies, disregarding word order and syntactic structure. For a document d with vocabulary V, the BoW vector is:

$$\mathbf{x} = [x_1, x_2, \ldots, x_{|V|}],$$

where x_i is the frequency of the i-th word in V. For example, consider the Vietnamese feedback "Giảng viên nhiệt tình hỗ trợ" with vocabulary $V = \{\text{giảng}, \text{viên}, \text{nhiệt}, \text{tình}, \text{hỗ}, \text{trợ}\}$. The BoW vector is:

$$\mathbf{x} = [1, 1, 1, 1, 1, 1],$$

as each word appears once.

Strengths: BoW is computationally efficient and simple, ideal for large datasets and baseline models.

Weaknesses: It ignores word order and semantic relationships, critical for Vietnamese due to its tonal nature (e.g., "tình" in "nhiệt tình" vs. "tình cảm" has context-dependent meanings), limiting its effectiveness for short, ambiguous feedback.

3.2 TF-IDF

Term Frequency-Inverse Document Frequency (TF-IDF) (Manning et al., 2008) enhances BoW by weighting terms based on their importance in a document relative to a corpus. The TF-IDF score for term t in document d within corpus D is:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \cdot \text{IDF}(t, D),$$

where:

- $\text{TF}(t, d)$: Frequency of term t in document d.
- $\text{IDF}(t, D) = \log\left(\frac{|D|}{\text{DF}(t,D)}\right)$, with $|D|$ as the number of documents and $\text{DF}(t, D)$ as the number of documents containing t.

Example: For the feedback "Giảng viên nhiệt tình hỗ trợ" in a corpus of 16,175 feedback instances, suppose "nhiệt tình" appears once in the feedback (TF $= 1$) and in 4,000 instances (DF $= 4,000$). Then:

$$\text{IDF} = \log\left(\frac{16,175}{4,000}\right) = \log(4) \approx 1.386,$$
$$\text{TF-IDF} = 1 \cdot 1.386 \approx 1.386.$$

Strengths: TF-IDF prioritizes rare, discriminative terms, improving performance over BoW in sparse, high-dimensional data.

Weaknesses: It neglects word order and semantic relationships, failing to capture tonal nuances in Vietnamese (e.g., "hỗ trợ" in positive vs. neutral contexts), limiting its ability to resolve sentiment ambiguity.

3.3 Tokenization

Tokenization (Jurafsky and Martin, 2023) splits text into tokens for processing by deep learning models. For Vietnamese, tokenizers like PhoBERT's handle word segmentation due to the language's non-space-separated nature.

Example: The title "Khuyến mãi công nghệ mới" is tokenized as:

[Khuyến, mãi, công, nghệ, mới].

Each token is mapped to an ID in the model's vocabulary.

Strengths: Enables precise text processing, critical for Vietnamese's tonal and syllabic structure.

Weaknesses: Errors in tokenization (e.g., splitting "khuyến mãi" incorrectly) can degrade model performance.

3.4 SMOTE

Synthetic Minority Oversampling Technique (Chawla et al., 2002) is a data augmentation method designed to address class imbalance by generating synthetic samples for the minority class. In our dataset, the Neutral class (698 entries) is significantly underrepresented compared to Positive (8,038) and Negative (7,439), potentially biasing models toward dominant classes. SMOTE mitigates this by interpolating new samples based on existing minority instances.

For a minority class sample $\mathbf{x}_i$, SMOTE selects one of its k-nearest neighbors $\mathbf{x}_{nn}$ (using Euclidean distance in the feature space) and generates a synthetic sample $\mathbf{x}_{\text{new}}$ as:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda \cdot (\mathbf{x}_{nn} - \mathbf{x}_i),$$

where $\lambda \in [0, 1]$ is a random interpolation factor.

Example: Consider a Neutral feedback entry "Giảng viên ổn" with its TF-IDF vector $\mathbf{x}_i = [0.1, 0.3, \ldots, 0.2]$. SMOTE selects a nearest neighbor, e.g., another Neutral feedback "Bài giảng bình thường" with vector $\mathbf{x}_{nn} = [0.15, 0.25, \ldots, 0.18]$, and generates a synthetic sample with $\lambda = 0.5$. The computation is:

$$\begin{aligned}
\mathbf{x}_{\text{new}} &= \mathbf{x}_i + \lambda \cdot (\mathbf{x}_{nn} - \mathbf{x}_i) \\
&= [0.1, 0.3, \ldots, 0.2] + \\
&= 0.5 \cdot ([0.15, 0.25, \ldots, 0.18] - \\
&= [0.1, 0.3, \ldots, 0.2]) \\
&= [0.1, 0.3, \ldots, 0.2] + \\
&= 0.5 \cdot [0.05, -0.05, \ldots, -0.02] \\
&= [0.125, 0.275, \ldots, 0.19].
\end{aligned}$$

This synthetic vector represents a new Neutral feedback, enhancing the minority class representation.

Strengths: SMOTE improves model performance on imbalanced datasets by increasing minority class samples, reducing bias toward majority classes. It is particularly effective for traditional models like Logistic Regression and SVM, which struggle with underrepresented classes.

Weaknesses: SMOTE may introduce noise if synthetic samples are generated in overlapping class regions, potentially confusing models. For Vietnamese feedback, where tonal nuances (e.g., "ổn" vs. "bình thường") are critical, SMOTE requires careful feature selection (e.g., TF-IDF or embeddings) to preserve semantic integrity.

3.5 Logistic Regression

Logistic Regression (Manning et al., 2008) is a linear model for classification, predicting the probability of a class using the sigmoid function. For a feature vector $\mathbf{x}$, the probability of class $y = 1$ (e.g., Positive) is:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w} \cdot \mathbf{x} + b)}},$$

where $\mathbf{w}$ is the weight vector and b is the bias term. **Example**: For the feedback "Giảng viên nhiệt tình", its TF-IDF vector $\mathbf{x}$ is used to predict the probability of Positive sentiment.

Strengths: Logistic Regression is simple, interpretable, and effective for linearly separable data.

Weaknesses: It struggles to capture complex, non-linear relationships in Vietnamese feedback, especially when tonal nuances affect sentiment.

3.6 Support Vector Machines (SVM)

Support Vector Machines (SVM) (Manning et al., 2008) classify text by finding the optimal hyperplane that maximizes the margin between classes. The optimization problem is:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i,$$

subject to:

$$y_i(\mathbf{w}\cdot\mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0,$$

where $\mathbf{w}$ is the weight vector, b is the bias, C is the regularization parameter, ξ_i are slack variables, y_i is the label (± 1), and $\mathbf{x}_i$ is the feature vector (e.g., TF-IDF).

Example: For the feedback "Giảng viên nhiệt tình", its TF-IDF vector is classified as Positive ($y_i = +1$) vs. Negative/Neutral ($y_i = -1$).

Strengths: SVM is robust for high-dimensional, sparse features like TF-IDF.

Weaknesses: It struggles with semantic ambiguity in short Vietnamese feedback and requires careful tuning of C.

3.7 Naive Bayes

Naive Bayes (Jurafsky and Martin, 2023) is a probabilistic classifier assuming feature independence. For a document with features $\mathbf{x} = [x_1, \ldots, x_n]$, the class probability is:

$$P(c|\mathbf{x}) \propto P(c)\prod_{i=1}^{n} P(x_i|c),$$

where $P(c)$ is the prior probability of class c, and $P(x_i|c)$ is the likelihood of feature x_i given class c.
Example: For the feedback "Giảng viên nhiệt tình", tokenized as {giảng, viên, nhiệt, tình}, suppose:

- $P(\text{Positive}) = 0.4$.
- $P(\text{nhiệt}|\text{Positive}) = 0.7$.
- $P(\text{tình}|\text{Positive}) = 0.6$.
- $P(\text{giảng}|\text{Positive}) = 0.5$.
- $P(\text{viên}|\text{Positive}) = 0.5$.

The score for Positive is:

$$\begin{aligned} P(\text{Positive}) &\cdot P(\text{giảng}|\text{Positive}) \cdot P(\text{viên}|\text{Positive}) \\ &\cdot P(\text{nhiệt}|\text{Positive}) \cdot P(\text{tình}|\text{Positive}) \\ &= 0.4 \cdot 0.5 \cdot 0.5 \cdot 0.7 \cdot 0.6 \approx 0.0042. \end{aligned}$$

Naive Bayes selects the class with the highest score.

Strengths: Fast and effective for small datasets.

Weaknesses: The independence assumption fails for context-dependent Vietnamese text, where words like "tình" rely on adjacent terms for sentiment.

3.8 XGBoost

XGBoost (Chen and Guestrin, 2016) is a gradient-boosting framework that builds an ensemble of decision trees to improve classification performance. It minimizes the loss function:

$$L = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K}\Omega(f_k),$$

where $l(y_i, \hat{y}_i)$ is the loss (e.g., log-loss for classification), $\Omega(f_k)$ is a regularization term, and f_k are decision trees.

Example: For the feedback "Giảng viên nhiệt tình", its TF-IDF features are fed to XGBoost to predict Positive sentiment.

Strengths: XGBoost captures non-linear relationships and is robust to noisy data.

Weaknesses: It requires careful feature engineering and hyperparameter tuning for text data.

3.9 Stacking Ensemble

Stacking Ensemble combines predictions from multiple base models (e.g., Logistic Regression, SVM, Naive Bayes, XGBoost) using a meta-learner (e.g., Logistic Regression) to improve accuracy. The process involves:

- Training base models on the dataset.
- Using their predictions as features for the meta-learner.

Example: For the feedback "Giảng viên nhiệt tình", predictions from base models are stacked to predict the final sentiment.

Strengths: Leverages complementary strengths of multiple models.

Weaknesses: Increases computational complexity and risks overfitting if not properly tuned.

3.10 PhoBERT

PhoBERT (Nguyen et al., 2020), based on BERT's transformer architecture, uses self-attention to model contextual relationships:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V},$$

where:

- $\mathbf{Q}, \mathbf{K}, \mathbf{V}$: Query, key, and value matrices.
- d_k: Dimension of keys.

Example: For "Thầy giảng bài dễ hiểu," PhoBERT's tokenizer splits it into tokens, and self-attention computes relationships (e.g., "Positive", "Negative" and "Neural"in context). Pre-trained on Vietnamese corpora, PhoBERT is fine-tuned for classification.

Strengths: Superior contextual understanding, ideal for tonal and short Vietnamese text.

Weaknesses: High computational cost and large memory requirements.

4 Experimental Settings

4.1 Dataset

Data Collection: The dataset comprises 16,175 Vietnamese student feedback entries from the Vietnam Student Feedback Data Set on Hugging Face, collected from educational platforms between January 2023 and March 2025. Approximately 8,000 feedback entries were gathered directly from course evaluation forms, with an additional 8,000 generated synthetically to address data scarcity and enhance diversity. Synthetic feedback was created using prompts like "Generate a Vietnamese student feedback for a [course/teacher] with [positive/negative/neutral] sentiment," ensuring alignment with real-world patterns.

Preprocessing: Feedback entries underwent cleaning to remove duplicates, special characters, and formatting inconsistencies (e.g., extra spaces). Text was normalized to lowercase, and Vietnamese diacritics were preserved to maintain linguistic integrity. Tokenization was performed using PhoBERT's tokenizer for deep learning models and underthesea for traditional models to handle Vietnamese word segmentation. For example, the feedback "Giảng viên nhiệt tình hỗ trợ" was tokenized as {giảng, viên, nhiệt, tình, hỗ, trợ}.

Annotation: Labels (Positive, Negative, Neutral) were assigned using keyword-based rules (e.g., "nhiệt tình" for Positive, "khó hiểu" for Negative), followed by manual validation by two annotators to ensure accuracy. Ambiguous cases (e.g., "Giảng viên ổn" as Neutral or Positive) were resolved through consensus. The dataset's class distribution is approximately balanced.

Challenges: The dataset exhibits severe class imbalance, with Neutral feedback (698 entries) significantly underrepresented compared to Positive (8,038) and Negative (7,439), potentially biasing models toward dominant classes. Short feedback texts and Vietnamese tonal nuances (e.g., "ổn" in Neutral vs. Positive contexts) complicate sentiment classification. Synthetic data, constituting half of the dataset, may introduce subtle biases despite manual validation efforts to align with real-world patterns.

Category	Number of Feedback Entries
Positive	8,038
Negative	7,439
Neutral	698
Total	16,175

Bảng 1: Dataset statistics, showing approximate distribution of student feedback entries across sentiment categories.

4.2 Baselines

To ensure a fair comparison, we implement baseline approaches for Vietnamese student feedback sentiment classification, leveraging traditional machine learning and deep learning methods:

- **Sentiment Classification with TF-IDF and BoW Features**: We implement traditional classifiers, including Logistic Regression, Support Vector Machines (SVM) (Manning et al., 2008), and Naive Bayes (Jurafsky and Martin, 2023), using two feature representations: Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF). These models are selected for their computational efficiency and effectiveness on sparse text data, such as student feedback like "Giảng viên nhiệt tình hỗ trợ."
- **Sentiment Classification with Deep Learning Models**: We evaluate PhoBERT (Nguyen et al., 2020), a BERT-based model pre-trained for Vietnamese. PhoBERT is fine-tuned to classify feedback into three sentiment categories: Positive, Negative, and Neutral, leveraging its ability to capture tonal and contextual nuances in Vietnamese text.

4.3 Settings

Our proposed framework consists of two core components:

- **Sentiment Classification with Traditional Machine Learning**: We treat sentiment classification as a multiclass classification task. Following standard approaches (Manning et al., 2008; Jurafsky and Martin, 2023), we employ four classifiers: Logistic Regression, SVM, Naive Bayes, and XGBoost (Chen and Guestrin, 2016). Each classifier is trained with BoW and TF-IDF features extracted from a dataset of 16,175 Vietnamese student feedback entries. Additionally, a Stacking ensemble combines predictions from these classifiers using a meta-learner (Logistic Regression) to enhance performance.

- **Sentiment Classification with Deep Learning**: We experiment with PhoBERT (Nguyen et al., 2020), fine-tuned using the Hugging Face Transformers library. PhoBERT leverages contextual embeddings to model semantic nuances in feedback, such as distinguishing "ổn" in Neutral vs. Positive contexts, addressing the challenges of short texts and class imbalance.

- **PyTorch** (Paszke et al., 2019): A deep learning framework used for training PhoBERT models.

- **Hugging Face Transformers** (Wolf et al., 2020): Provides pre-trained PhoBERT models and supports fine-tuning.

- **Scikit-learn** (Pedregosa et al., 2011): Used to implement traditional machine learning classifiers (SVM, Naive Bayes).

4.4 Evaluation Metrics

The sentiment classification task is evaluated using the following metrics:

- **Sentiment Classification**: We use accuracy, precision, recall, and weighted F1-score to account for severe class imbalance across the three sentiment categories (Positive, Negative, Neutral), as described in (Manning et al., 2008). The weighted F1-score is particularly crucial given the underrepresentation of Neutral feedback (698 entries) compared to Positive (8,038) and Negative (7,439).

5 Results and Discussion

We evaluated the performance of traditional machine learning models—Logistic Regression (LR), Support Vector Machines (SVM), Naive Bayes (NB), XGBoost (XGB), and Ensemble Learning (En-L)—using Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) representations on the task of classifying 16,175 Vietnamese student feedback entries into Positive, Negative, and Neutral sentiments. Additionally, we integrated PhoBERT, a pre-trained Vietnamese transformer model, as a state-of-the-art baseline. The experiments were conducted in two phases: first with default hyperparameters, then with hyperparameters optimized using Random Search. Performance was assessed using accuracy, precision, recall, and weighted F1-score to address the severe class imbalance, particularly the underrepresentation of the Neutral class (698 entries).

5.1 Default Hyperparameters Results

Table 2 presents the performance of all models using default hyperparameters. The results show that ensemble learning consistently outperformed individual models across both feature representations. With BoW features, the ensemble method achieved 90.07% accuracy and 90.88% F1-score, while XGBoost performed second-best with 89.74% accuracy. For TF-IDF features, the ensemble method reached the highest performance with 91.51% accuracy, though notably showing a slight decrease in F1-score to 90.01%.

Feature	Models	Acc.	Pre.	Re.	F1
BOW	LR	86.00	88.14	85.99	86.91
	SVM	88.75	90.39	88.75	89.41
	NB	84.71	84.86	84.71	84.77
	XGB	89.74	89.72	89.74	89.73
	En-L	90.07	90.80	90.07	90.88
TF-IDF	LR	85.49	90.31	85.49	87.72
	SVM	85.86	89.40	85.86	87.28
	NB	82.11	86.24	82.11	83.83
	XGB	87.51	90.33	87.51	88.66
	En-L	**91.51**	**91.09**	**91.51**	**90.01**

Bảng 2: Performance comparison with default hyperparameters

5.2 Hyperparameter Optimization Results

Table 3 shows the results after applying Random Search for hyperparameter optimization. The optimization process significantly improved several models' performance. Most notably, Logistic Regression with BoW features improved from 86.00% to 88.87% accuracy, and XGBoost with TF-IDF features increased from 87.51% to 89.82% accuracy. The ensemble learning method maintained its superior performance, achieving 91.51% accuracy with TF-IDF features, consistent with the default hyperparameter results.

Feature	Models	Acc.	Pre.	Re.	F1
BOW	LR	88.87	90.25	88.87	89.45
	SVM	88.75	90.39	88.75	89.41
	NB	84.71	84.86	84.71	84.77
	XGB	89.74	89.72	89.74	89.73
	En-L	90.97	89.72	89.74	89.73
TF-IDF	LR	86.29	89.47	86.27	87.57
	SVM	85.86	89.40	85.86	87.28
	NB	82.11	86.24	82.11	83.83
	XGB	89.82	90.36	89.82	90.08
	En-L	**91.51**	**91.09**	**91.51**	**90.01**

Bảng 3: Performance comparison with optimized hyperparameters using Random Search

5.3 PhoBERT Performance

To establish a state-of-the-art baseline, we fine-tuned PhoBERT, a pre-trained Vietnamese transformer model, on our dataset. PhoBERT achieved remarkable performance with 95% accuracy, significantly outperforming all traditional machine learning approaches. This superior performance demonstrates the effectiveness of transformer architectures in capturing complex semantic relationships and contextual nuances in Vietnamese sentiment analysis.

5.4 Analysis and Discussion

The experimental results reveal several key insights about Vietnamese sentiment analysis performance:

Transformer vs. Traditional ML Performance Gap: PhoBERT's 95% accuracy represents a substantial improvement over the best traditional approach (ensemble learning with 91.51% accuracy), highlighting a 3.49 percentage point gap. This performance difference underscores the advantages of pre-trained language models in understanding Vietnamese linguistic nuances, including context-dependent sentiment expressions, implicit meanings, and cultural references that traditional feature-based methods struggle to capture.

Feature Representation Impact on Traditional Models: Among traditional approaches, TF-IDF consistently outperformed BoW across most model configurations, particularly for ensemble learning methods. This suggests that term weighting and document frequency normalization are crucial for capturing sentiment patterns in Vietnamese text when deep contextual understanding is not available.

Traditional Model Performance Hierarchy: Within the traditional ML paradigm, ensemble learning emerged as the top-performing approach, achieving 91.51% accuracy with TF-IDF features. XGBoost consistently ranked second across different configurations, demonstrating the effectiveness of gradient boosting for this classification task. Traditional methods like Naive Bayes showed the lowest performance, possibly due to its strong independence assumption not holding well for sentiment-laden text features.

Hyperparameter Optimization Effects: Random Search optimization showed mixed results across traditional models. While some models like Logistic Regression benefited significantly (2.87% accuracy improvement with BoW), others like SVM showed minimal gains. This suggests that some algorithms are more sensitive to hyperparameter tuning in Vietnamese sentiment analysis tasks.

Class Imbalance Handling: Interestingly, PhoBERT's superior performance may partially stem from its better handling of the class imbalance problem. The transformer's attention mechanism and contextual understanding likely provide better discrimination for the underrepresented Neutral class compared to traditional feature-based approaches.

Computational Efficiency vs. Performance Trade-off: While PhoBERT achieves the highest accuracy, it requires significantly more computational resources and training time compared to traditional methods. For applications with strict resource constraints, XGBoost with optimized hyperparameters (89.82% accuracy) still represents a viable alternative, offering reasonable performance with much lower computational overhead.

6 Conclusion

This study comprehensively evaluated traditional machine learning approaches alongside PhoBERT for Vietnamese student feedback sentiment analysis using 16,175 feedback entries. Our comparative analysis established clear performance benchmarks across different methodological approaches.

6.1 Key Achievements

The research successfully demonstrates the effectiveness of various approaches for Vietnamese sentiment analysis:

- **Performance Hierarchy:** PhoBERT achieved the highest accuracy at 95%, followed by ensemble learning with TF-IDF (91.51%), and optimized XGBoost (89.82%).
- **Feature Representation Insights:** TF-IDF consistently outperformed BoW across traditional models, proving crucial for Vietnamese text sentiment analysis.
- **Optimization Impact:** Random Search hyperparameter tuning showed variable benefits, with some models (Logistic Regression) improving significantly while others showed minimal gains.
- **Practical Solutions:** Established a spectrum of deployment options balancing accuracy and computational efficiency for different resource constraints.

6.2 Limitations of Traditional Machine Learning Approaches

Despite achieving reasonable performance, traditional ML methods face several inherent limitations in Vietnamese sentiment analysis:

- **Contextual Understanding:** Traditional feature-based methods struggle with Vietnamese context-dependent sentiment expressions, implicit meanings, and cultural nuances that significantly impact sentiment interpretation.
- **Class Imbalance Handling:** The 3.49% performance gap with PhoBERT suggests traditional methods are less effective at handling the underrepresented Neutral class, likely due to insufficient contextual discrimination capabilities.
- **Feature Engineering Dependency:** Traditional approaches require extensive manual feature engineering and domain expertise, limiting their adaptability to evolving Vietnamese language patterns and educational contexts.
- **Semantic Representation Limitations:** BoW and TF-IDF representations cannot capture complex semantic relationships, word order dependencies, and long-range contextual information crucial for accurate sentiment analysis.

6.3 Future Directions

- **Hybrid Architectures:** Develop combinations of PhoBERT feature extraction with lightweight traditional classifiers to balance accuracy and computational efficiency.
- **Domain-Specific Optimization:** Fine-tune PhoBERT with educational Vietnamese text and explore continued pre-training strategies to further improve performance.
- **Real-world Deployment:** Investigate model compression, knowledge distillation, and efficient inference techniques to make high-performance sentiment analysis accessible in resource-constrained educational environments.
- **Cross-domain Generalization:** Extend evaluation to diverse Vietnamese educational contexts and develop domain adaptation techniques for broader applicability.

This work establishes important benchmarks for Vietnamese educational sentiment analysis, demonstrating both the potential and limitations of traditional ML approaches while highlighting the significant advantages of modern transformer-based methods for this challenging NLP task.

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